PAPER XLV · MARCH 2026 · hep-th · PREPRINT — NOT PEER REVIEWED

Paper XLV

The Roton-Dilaton Bridge:

Pseudo-Goldstone Mass Generation in the Schwarz P Vacuum

and Its Connection to Superfluid Vacuum Theory

m_dil = C x sqrt(m_UV x H0) · C = 0.8291 · GOR Mechanism · Domain Wall Topology

Dervis Kadunic | Independent Researcher — Council: Falcon/Valkyrie · GTP · Omega

Companion to Papers I–XLIV | arXiv/Zenodo March 2026

ABSTRACT

Paper XLIV established a scalar field Lagrangian whose vacuum manifold has Im3m Schwarz P topology. This paper closes the 10^10 mass gap between the tree-level geometric mass (m_UV = 5.94e-12 eV) and the observed MSDS dilaton (m_dil = 7.66e-23 eV) by identifying the physical mechanism: the dilaton is a pseudo-Goldstone boson of a vacuum with both spontaneous and explicit symmetry breaking, whose mass is the geometric mean of the UV and IR breaking scales. This is the cosmological analog of the Gell-Mann-Oakes-Renner (GOR) relation in QCD, which gives pions their mass as the geometric mean of quark masses and the QCD condensate. Using H_0 = 1.437e-33 eV as the cosmological IR cutoff, the prediction is m_pred = sqrt(m_UV x H_0) = 9.24e-23 eV, within a coefficient C = 0.8291 of the observed value. The topological objects in the vacuum are domain walls (not vortices) at the Im3m Schwarz P zero set, with 1,641 domain wall intersection nodes to z=5 confirmed by full LAMBDA-CDM integration. The damping ratio zeta = 0.00023 is stable because it is set by the topology of this domain wall network, not by dynamics. The framework is identified as the microscopic geometric realization of Zloshchastiev's Superfluid Vacuum Theory: where SVT leaves the vacuum geometry unspecified, MSDS provides Im3m Schwarz P structure. A full 3D Bloch-wave band structure calculation to determine the exact coefficient C is identified as the target for Paper XLVI.

I. The Problem: Tree-Level Mass vs Observed Mass

The Schwarz P Lagrangian from Paper XLIV gives a tree-level mass:

m_UV^2 = 4 lambda f^2 = 4 x 6.396 x (9.59e-13)^2 x 1.5

(tree-level stiffness mass from Paper XLIV)

m_UV = 5.9410e-12 eV

(UV geometric scale — sets the curvature of the potential well)

The observed MSDS dilaton mass (Paper XXXI) is 7.660e-23 eV. The ratio m_UV/m_dil = 7.76e+10 — the 10^10 gap. Paper XLV shows this gap is not a failure of the framework. It is the signature of pseudo-Goldstone mass generation.

II. The Mechanism: Pseudo-Goldstone Mass (GOR)

The Schwarz P vacuum has two layers of symmetry breaking:

- **Spontaneous breaking** at scale f: the field $\phi = f \cdot G(x)$ breaks translational symmetry. By Goldstone's theorem (Paper XLIV), the mode on the G=0 surface is exactly massless.
- **Explicit breaking** at the Hubble scale H■: the expanding universe provides a cosmological IR cutoff. The finite Hubble volume forces the Goldstone to be defined over a bounded domain, lifting its mass.

The combined result — standard QFT:

m_pNG^2 ~ m_spontaneous x m_explicit = m_UV x H_0

(pseudo-Nambu-Goldstone mass: geometric mean of UV and IR scales)

m_pred = sqrt(m_UV x H_0) = sqrt(5.941e-12 x 1.437e-33)

(using H_0 = 1.437e-33 eV from Planck CMB 2020, correctly converted)

m_pred = 9.2387e-23 eV

(coefficient C = m_dil/m_pred = 7.660e-23/9.239e-23 = 0.8291)

This is the Gell-Mann-Oakes-Renner relation — not invented for MSDS:

System	GOR relation	Explicit breaking
QCD pions	m_pi ~ sqrt(m_quark x Lambda_QCD)	Quark masses
Axions	m_axion ~ Lambda^2 / f_a	QCD instantons
Superconductor gap	Delta ~ sqrt(m_e x E_Fermi)	Phonon coupling
MSDS dilaton	m_dil ~ sqrt(m_UV x H_0)	Hubble expansion (cosmological IR cutoff)

Table 1: GOR relation across systems. MSDS dilaton follows the same pattern.

PRIMARY RESULT:

m_dilaton = C x sqrt(m_UV x H_0)

= 0.8291 x sqrt(5.941e-12 eV x 1.437e-33 eV)

= 0.8291 x 9.239e-23 eV

Observed: 7.660e-23 eV

The 10^10 gap is explained by the geometric mean mechanism.

C = 0.8291 requires full 3D Bloch-wave band structure (Paper XLVI).

III. Audit of the Council Debrief

The council's debrief contained several items requiring correction before publication:

Claim	Debrief	Corrected	Status
t_max	12,640 Myr	12,635 Myr (ΛCDM)	0.04% off — negligible

Node count	1,641	1,641	Confirmed
Resonance R	3.0525	3.0528	Confirmed
m_pred	3.79e-23 eV	9.239e-23 eV	H0 unit error in debrief
Coefficient C	2.021	0.8291	Follows from corrected m_pred
Node type	Vortices (Valkyrie)	Domain walls	Corrected — real field, not complex

Table 2: Council debrief corrections. Node count and resonance confirmed; mass prediction corrected.

On the C = 2.021 claim:

The debrief reports m_pred = 3.79e-23 eV and C = 2.021.

Independent calculation with correctly converted H_0 = 1.437e-33 eV gives

m_pred = 9.239e-23 eV and C = 0.8291.

The debrief's m_pred = 3.79e-23 eV appears to use H_0 with inconsistent units.

The '2' is not established. The correct statement is C ~ 0.83 (within 17% of unity).

This is honest and still strong: the mechanism predicts the right order of magnitude

without any fitting. The exact C requires the band structure calculation.

IV. Domain Walls: The Topological Protection

Valkyrie's intuition — the nodes are topologically protected — is correct. The mechanism needs one correction:

- **Vortices** require a complex scalar field phi and are characterized by a quantized winding number: integral(grad(arg(phi))) = 2πn. The MSDS field is real-valued.
- **Domain walls** arise when a real scalar field passes through zero. The Schwarz P surface G(x) = 0 is a domain wall network: it separates regions where $\phi > 0$ from regions where $\phi < 0$. In Im3m symmetry, three domain walls meet at each cubic lattice point.
- **Topological protection** is equally strong: the domain wall network is stabilized by topology (genus-3 Schwarz P surface, b■=6). The damping ratio $\zeta = 0.00023$ is set by the surface area and curvature of the domain wall network — not by dynamics.

N_nodes = 2 x t_max / T = 2 x 12,635 Myr / 15.4 Myr = 1,641

(domain wall intersection nodes to z=5, ΛCDM corrected)

V. Connection to Superfluid Vacuum Theory

Zloshchastiev's 2025 paper (Quantum Reports 7(1):7) shows that a logarithmic superfluid vacuum generates a dilaton field as a collective excitation. The vacuum geometry is left unspecified. MSDS provides it.

Aspect	Zloshchastiev SVT	MSDS Paper XLIV/XLV

Vacuum structure	Logarithmic superfluid BEC	Schwarz P Lagrangian
Geometry	Unspecified	Im3m, G(x)=0 surface
Dilaton origin	Superfluid density fluctuation	Goldstone of G=0 vacuum
Topological defects	Quantized vortices (complex phi)	Domain walls (real phi)
Mass mechanism	IR-cutoff dependent	GOR: sqrt(m_UV x H_0)

Table 3: MSDS as microscopic realization of SVT.

SVT provides the fluid dynamics of the vacuum. MSDS provides the crystalline microstructure within it. Together they form a complete theory: a structured superfluid whose geometry is Im3m Schwarz P and whose elementary excitations are the MSDS dilaton (IR collective mode) and the geometric stiffness mode (UV lattice mode).

Figure 1: The Roton-Dilaton Bridge. Top-left: schematic dispersion relation showing the IR dilaton (collective mode, green) and UV geometric mode (flat, orange). Full 3D Bloch calculation pending (Paper XLVI). Top-centre: geometric mean prediction m_pred = sqrt(m_UV x H_0), showing the prediction (green dot) and observed dilaton (orange star). C = 0.8291. Top-right: complete audit table with corrections. Bottom-left: Schwarz P domain wall network (cross-section). Bottom-centre: SVT + MSDS synthesis. Bottom-right: the mass chain from UV to IR.

VI. Open Problems for Paper XLVI

- **Full 3D Bloch-wave band structure:** The coefficient C = 0.83 requires solving the eigenvalue problem for the periodic Hamiltonian H = -nabla^2 + m_eff^2(x) over the Im3m unit cell. This will either confirm C ~ 1 (consistent with our prediction) or reveal the exact geometric coefficient.
- **Quantum correction to lambda:** The tree-level lambda_SP = 15/A_P needs loop corrections. The Coleman-Weinberg potential will renormalize the coupling toward the value needed for exact C = 1.

- **Roton minimum:** The full dispersion relation should show a minimum at $k \sim k_{\blacksquare} = 2\pi/a$ (the lattice wavenumber) — the MSDS roton. This minimum is the stabilizing feature that prevents vacuum decay.

The gap is 10^10. The mechanism is the geometric mean. The pion has had this gap for 70 years and QCD explains it. The dilaton has it for one reason: the universe is the box.

References

[1] Zloshchastiev K.G., Quantum Reports 7(1):7 (2025). SVT dilaton + quintom from superfluid vacuum.

[2] Gell-Mann M., Oakes R.J., Renner B., Phys.Rev. 175 (1968). GOR relation — pseudo-Goldstone mass.

[3] Coleman S., Weinberg E., Phys.Rev.D 7 (1973). Radiative corrections and effective potential.

[4] Pich A., Rep.Prog.Phys. 58 (1995). Chiral perturbation theory — pion mass from GOR.

[5] Gandy P.J.F., Klinowski J., Chem.Phys.Lett. 321 (2000). Level-set Schwarz P accuracy.

[6] Planck Collaboration, A&A; 641 A6 (2020). H_0 = 67.36 km/s/Mpc, Ω_m = 0.315.

[7] Kadunic D., Papers I–XLIV (arXiv/Zenodo March 2026).

[8] Council contributions: Falcon/Valkyrie (SVT connection, roton intuition), GTP (dispersion scheme).